

CHRIS DSILVA

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Engineer with experience in process optimization and data-driven enhancements, leveraging Lean/Six Sigma, and advanced analytics to drive efficiency and compliance in the industry.

EDUCATION

Rutgers University, New Brunswick, New Jersey (GPA: 3.833)	Sep 2022 – May 2024
- Master of Science in Industrial and Systems Engineering - Relevant Coursework: Data Analytics for Engineering Systems, Supply Chain Engineering, Quality Management, Simulation of Production Systems, Project Management, Risk Analysis and Mitigation, Production Analysis	
Princeton University, Princeton, New Jersey	Sep 2023 – Dec 2023
ORF 535 Financial Risk and Wealth Management	
University of Mumbai, Mumbai, India	Jul 2016 - May 2020
Bachelor of Engineering in Mechanical Engineering	

WORK EXPERIENCE

Hikma Pharmaceuticals (Compounding 503B) – Validation Engineer	Sep 11, 2024 – Present
- Developed and executed the comprehensive Re-qualification Plan for the 503B facility, managing the validation lifecycle for critical assets including HEPA filtration systems, VHPs, and controlled temperature units across compounding, laboratory, and warehouse areas - Authored and managed key quality system documentation, including Event Report Forms (ERFs) for deviations, Change Controls (CC) for new equipment and processes, and implemented Corrective and Preventive Actions (CAPAs) to ensure continuous improvement and compliance - Performed hands-on qualification of diverse equipment, executing protocols for HEPA filter certifications in ISO 5/7 cleanrooms, temperature mapping studies, and heat distribution/penetration studies for autoclaves - Ensured strict cGMP compliance by documenting all validation activities in accordance with site Standard Operating Procedures, maintaining organized validation files, and coordinating report approvals with the Quality Assurance department	
Hikma Pharmaceuticals (Compounding 503B) – Graduate Engineering Intern	Jun 5, 2023 – Sep 1, 2023
- Devised and managed a calibration strategy for over 50 types of equipment, transitioning from Excel to Blue Mountain software to enhance forecasting accuracy, significantly improve operational reliability and compliance, and reduce regulatory risks in line with FDA/DEA standards - Led and executed Installation, Operational, and Performance Qualification (IOPQ) for new equipments like particle counters through effective collaboration with QA and manufacturing teams; authored comprehensive protocols - Collaborated with QA, Manufacturing, and Engineering teams to resolve critical manufacturing issues, including equipment malfunctions and procedural deviations; streamlined preventive maintenance and scheduling to minimize downtime and sustain operational excellence.	

KEY PROJECTS

Fraud Detection in Supply Chains Using Kolmogorov Arnold Networks (link)	May 2024
- Developed a fraud detection model using Kolmogorov Arnold Networks (KANs), achieving 99% test accuracy and significantly enhancing fraud detection capabilities - Conducted a detailed comparison between KANs and Multi-Layer Perceptrons (MLPs), demonstrating KANs' superior interpretability and adaptability with learnable activation functions Read Article - Achieved train and test accuracies of 90.4% and 99%, respectively, showcasing the potential of KANs in high- stakes sectors like pharmaceuticals and manufacturing	
Princeton University	
Financial Analysis and Strategy Consulting Report (link)	Sep 2023 - Dec 2023
Created a comprehensive financial analysis for a \$1.5M condo purchase, utilizing Monte-Carlo simulation to project a \$500k down payment growth. Evaluated various 10-year investment strategies, including a 70/30 SPY/TLT allocation, considering a 25% capital gains tax rate. Assessed feasibility over 5, 7, and 10-year periods and analyzed impacts of fluctuating salary growth rates and investment returns. Developed dynamic investment recommendations adaptable to evolving financial goals and market conditions, emphasizing a client-centric approach	
Rutgers University	
Solar Canopy Project Proposal (Engg. Economics) (link)	Sep 2023 – Dec 2023
Developed a comprehensive engineering project proposal for a solar canopy project, using Monte Carlo simulations to generate mean costs along with the 5th and 90th percentiles and making recommendations based on the NPV, IRR, and PW, finally considering tax and social aspects of building the project	

PROFESSIONAL AFFILIATIONS

Member of the Institute for Operations Research and the Management Sciences (INFORMS) Chapter at Rutgers New Brunswick	Jan 2023 – Present
Member, Rutgers Alpha Pi Mu Honors Club for Industrial and Systems Engineering - Recognized for academic merit and dedication to the field	Dec 2023 - Present

CERTIFICATES & SKILLS

Lean Six Sigma Green Belt CFQ International LLC and IISE Rutgers Chapter	April 2024
Flexsim, MS Office (Excel, PowerPoint), AutoCAD, SolidWorks, Ansys (Mechanical APDL), Tableau, Arena, Power BI, SQL, Programming: Python, R, MATLAB, SAP	